Capstone Project Proposal

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Title of Capstone Project

Predicting Station Imbalance Risk in Bike Share Toronto Based on Weather and Large-Scale Event Scenarios.

Project Idea

Nowadays bike-sharing systems are an important component of sustainable urban transportation in cities. Although, the effectiveness of these systems depends on maintaining an appropriate balance of bicycles across stations, the station imbalance is one of the most significant operational challenges faced by bike-sharing systems. Nevertheless, there are cases where stations become either completely full or empty, hindering users from renting or returning bicycles.

This Capstone project is focusing on applying advanced data analytics techniques to predict station imbalance risk in Bike Share Toronto system by analyzing demand variability patterns and examining how data-driven methods can improve the efficiency and sustainability of bike-sharing systems. The project seeks to identify high-risk stations and time periods where imbalance is most likely to occur by exploring how predictive models can be applied to simulate extreme demand scenarios and, by leveraging historical bike-sharing data, real-time station information, weather data, and records of large-scale public events such as the FIFA World Cup 2026.

Problem Identification

The city of Toronto, being so large and having four seasons throughout the year, among other external conditions, faces many difficulties in managing all its systems, which includes Toronto Bike Share. This bike loan system is spread across more than 1,000 stations and more than 10,000 bikes, and yet it still has stations with imbalances, where the system is less reliable. All of this negatively affects the user experience and automatically increases operating costs due to poor system optimization.

Large-scale public events, such as festivals, concerts, and international sporting events, can create sudden and localized increases in demand, which often intensify station imbalance. As Toronto prepares to host matches during the FIFA World Cup 2026, understanding how these high demand situations may impact bike-sharing availability has become increasingly important.

This project focuses on a real operational challenge by moving away from reactive, historically approaches and instead applying predictive and scenario-based analytics to improve preparedness for both normal and exceptional demand conditions.

Research Question

How can temporal, spatial, weather-related, and large-scale event factors be used to predict station imbalance risk in Bike Share Toronto?

Hypothesis

Station imbalance risk in Bike Share Toronto is significantly influenced by time-related patterns, station location, weather conditions, and the presence of large-scale public events. Predictive models that incorporate these factors can anticipate high-risk imbalance scenarios, including extreme demand conditions similar to those expected during the FIFA World Cup 2026, more accurately than models based solely on historical averages.

Data Sources

This project will rely on real, publicly available, and reproducible datasets, including:

1. Bike Share Toronto – GBFS Data

- Station Information (station location, capacity)
- Station Status (real-time bike and dock availability)
- Accessed through publicly available GBFS endpoints

2. Historical Bike Share Toronto Usage Data
    - Trip-level historical data published via the City of Toronto Open Data Portal
    - Used to derive station-level demand and imbalance metrics
3. Weather Data (Environment and Climate Change Canada, n.d.)
    - Hourly historical and daily weather variables (temperature, precipitation, wind)
    - Retrieved from a free public weather API
4. Large-Scale Public Event Data (City of Toronto, 2026)
    - Curated dataset of major public events (festivals, concerts, city-wide celebrations)
    - Event dates and approximate locations used to model demand shocks
    - Historical events will serve as proxies for scenario-based analysis related to the FIFA World Cup 2026

Proposed Methodology

The project will follow a structured data analytics lifecycle:

1. Data Collection and Integration
    - Automated ingestion of historical and near real-time bike-share data
    - Integration with weather data and event indicators

2. Exploratory Data Analysis (EDA)
    - Analysis of demand patterns, peak usage periods, and station imbalance occurrences
    - Comparative analysis between normal days and event-driven days
3. Feature Engineering
    - Temporal features (hour of day, weekday/weekend, seasonality)
    - Station-level features (capacity, spatial clustering)
    - Weather-related features
    - Event-based indicators (event day, event intensity, proximity to stations)
4. Predictive Modeling
    - Development of baseline models using historical averages
    - Application of machine learning models (e.g., logistic regression, random forest) to predict station imbalance risk
    - Evaluation using appropriate performance metrics
5. Model Validation and Comparative Analysis
    - Comparison between baseline approaches and predictive models
    - Assessment of model robustness using historical large-scale events as high-demand scenarios

- Scenario-based evaluation to approximate conditions similar to those expected during the FIFA World Cup 2026

6. Visualization and Insights
    - Development of dashboards highlighting high-risk stations and time periods
    - Clear communication of findings and operational insights

Expected Outcomes and Contributions

The expected outcomes of this project include:

- Identification of key drivers contributing to station imbalance risk
- Predictive models capable of anticipating imbalance under normal and high-demand conditions
- Comparative evidence demonstrating the added value of weather and event-driven features
- Scenario-based insights supporting planning and preparedness for large international events such as the FIFA World Cup 2026

This project contributes a practical, data-driven framework for improving the reliability and resilience of urban bike-sharing systems under both routine and extreme demand conditions.

Conclusion

By integrating historical bike-share data, real-time station information, weather variables, and large-scale event scenarios, this Capstone project addresses a relevant and operationally meaningful challenge in urban transportation while providing actionable insights for smart city mobility planning.

References

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